

M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo

Full recording stitch

[Open full recording](#)

Clock orbit: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right · Effective 3× speed (video encoded at 1.5×, player at 2×). [Open orbit comparison](#)

Clock task: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right. Task replay uses the latest orbit mesh under the temporary demo policy. [Open task comparison](#)

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

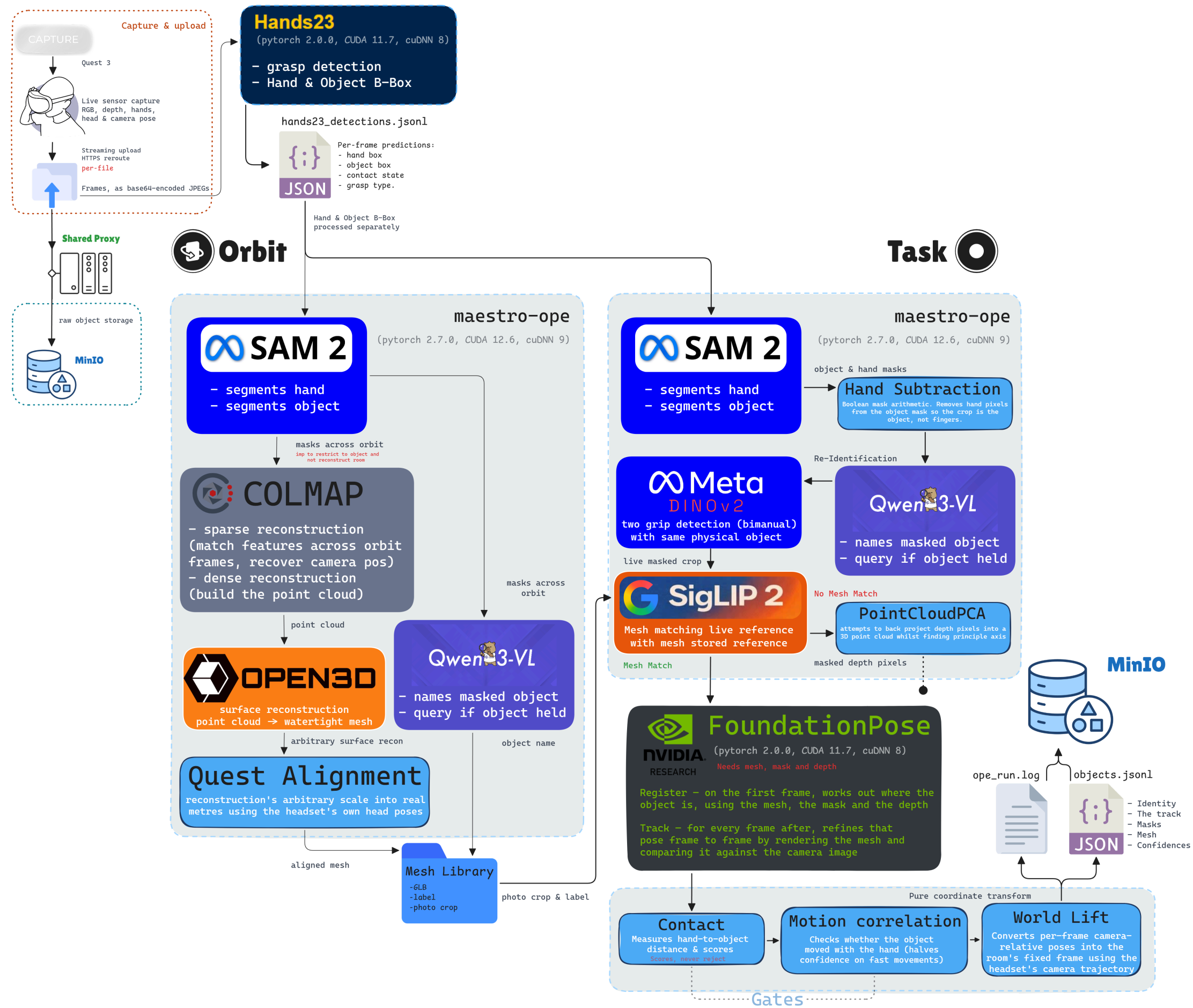


Figure 1. Current Maestros VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

1

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

[Quest sensor capture](#)

2

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

[MinIO object storage](#)

3

Hands23 hand detection

Provides hand/object contact evidence and candidate boxes before object processing begins.

[Hands23 code and explanation](#)

4

SAM2 segmentation

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

[SAM2 overview and code](#)

5a

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

[Project pose measurements](#)

5b

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

[FoundationPose overview and code](#)

6

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

[Qwen3-VL-4B-Instruct model](#)

7

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

[DINOv2 ViT-S/14 model](#) · [SigLIP base model](#)

8

COLMAP mesh reconstruction

Reconstructs object meshes from orbit capture frames for canonical mesh-backed replay.

[COLMAP tutorial and code](#)

9

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

[Unity XR Interaction Toolkit](#)

System Status

Confirmed steps come first, followed by demo patches, partial fixes and disabled work, then outcomes still to confirm. Open up to two cards at once. The side connections show where a working step still depends on unfinished follow-up.

Hands23: the starting point

A detected grip supplies the hand and object boxes that seed the rest of the pipeline.

Module 3 **Confirmed**

The live worker checks the separate Hands23 service and saves its detections before object processing. Missing detections now raise an error, instead of being mistaken for a recording with no object. A brief deliberate grip seeds the orbit route; task recordings use contact evidence to start tracking.

These boxes changed the architecture: SAM2 can now isolate both the hand and the object, so a separate SAM3 hand detector is no longer needed. A box is the starting region, not a finished mask or mesh.

[Hands23 code and explanation](#)

Grip-first orbit mesh generation

The reconstruction route produces real-scale meshes when a grip is detected and the orbit has usable visual detail.

Modules 3, 4, 8 **Confirmed**

Grip first, then slowly circle the object for the rest of the clip. Hands23 identifies the starting object; SAM2 follows its outline. Masked COLMAP matches visible features across the images and reconstructs a 3D point cloud. Open3D builds the surface, and Quest head poses set its scale in metres.

Candle and clock reconstructions are measured successes. This confirms the reconstruction method; it does not establish unattended upload-to-mesh automation for every session. The latest clock reconstruction was run and registered manually.

Route	Input and purpose	Project finding
Hunyuan3 D-Omni	One reference image; generates plausible hidden surfaces	Poor remote-control result despite a clean photograph.
TRELLIS. 2	One reference image; generates a textured mesh	Cleaner remote-control result; unseen surfaces and orientation still need checking.
Fuse3D	Several real views fused through a generative model	Strong front, thinner sides in the same comparison.
COLMAP + Open3D	Matches real image features across an orbit; reconstructs points, then a surface	Measured metric candle and clock meshes. Low-texture objects can still fail.

The earlier comparison is a small project trial, not a general model ranking. COLMAP is the measured orbit route, but the bottle and rolling pin failed at feature matching. The bowl had healthy pairwise matches and failed later in multi-view consistency.

Follow-ups: [mesh selection and replay](#) · [orientation and interaction](#).

[COLMAP tutorial and code](#)

FoundationPose with a trusted mesh

The mesh-backed service has run successfully; reliable tracking through hand occlusion still needs care.

Module 5b **Confirmed**

FoundationPose was already implemented. The recorded activation blockers were the wrong GPU being almost full from another job, and a mesh loader expecting OBJ while registered assets used

GLB. Moving the service to a GPU with room and supplying the expected mesh enabled real registration and more than 100 successful tracking calls.

The bottle trial still had 9 large rotation jumps in 112 transitions, mostly flip-and-recover pairs. Hand occlusion was a likely cause, not a proven diagnosis. A working container does not prove perfect tracking.

Default live task processing remains model-free while mesh matching is disabled. Borrowing a mesh for rendering does not rerun FoundationPose.

Without a trusted mesh, the remaining fallback issue is [PointCloudPCA](#).

[FoundationPose overview and code](#)

SAM3 removal: SAM2 is sufficient for the current prompts

Hands23 supplies the boxes; SAM2 creates the object and hand masks.

Modules 3, 4 Confirmed

The earlier reference-image mesh route needed hand pixels removed before generating a mesh. SAM3 was also used for an object-presence check. It was removed after the Hands23-guided SAM2 route was evaluated; this was more than removing a duplicate model.

Model	Plain-language role	Why this pipeline changed
SAM2	Select a region using a point or box, then follow its pixels through video.	Hands23 supplies an object box and a hand box. SAM2 makes separate object and hand masks; hand pixels are subtracted.
SAM3	Find and segment concepts using text or example prompts.	Previously used for hand removal and an object-presence check. Its hand mask failed on a real bottle recording; the presence score did not reliably distinguish a real grasp from a fake one.

SAM2 runs with two different prompts, producing two masks. Qwen performs the remaining held-object check. This keeps the pipeline simpler without claiming SAM3 is inferior for every task.

[SAM2 overview](#) · [SAM3 overview](#)

Episode naming endpoint

Names are built from existing object labels and capture dates, then served to Unity by session UUID.

`{object_key}_{counter}_{YYMMDD}`

The server uses existing Qwen3-VL labels and, when available, canonical category or label data. It normalises supported terms, rather than asking another model to invent a task title. The counter starts at 00 for each object-key/date pair and grows beyond 99. The capture date prefers device-local time.

Outcome	Example display name	Meaning
Waiting / processing	<code>unidentified_object_00_261005</code>	Pending; the object is not yet identified.
Finished, no object	<code>unidentified_object_00_261005</code>	Unidentified; distinct processing state even though the name has the same form.
One recognised object	<code>clock_00_261005</code>	A supported token from the existing label or canonical record.
Two different objects	<code>bottle.clock_00_261005</code>	Distinct tokens sorted alphabetically and joined with a dot.
Repeated object type	<code>clock_x2_00_261005</code>	Two tracked records with the same token.
Three different types	<code>bottle.candle.clock_00_261005</code>	Up to three distinct tokens remain explicit.
Four or more types	<code>multiobj4_00_261005</code>	Number counts distinct types, rather than total objects.
Unsupported label	<code>unidentified_object_00_261005</code>	A detected object can still have an unsupported name; check the state and label.
Device legacy title	<code>untitled</code>	Old device title, not a generated server naming outcome.

`GET /sessions/{uuid}/metadata` includes the display name, processing state, revision, timestamp and grasp result. Pending and finished-with-no-object remain distinct. Device folders are not renamed.

Mesh selection and headset replay

Meshes can render in replay; the current task selection policy is a temporary shortcut.

Modules 7, 9

Fixed for demo

- **Rendering:** demo sessions had meshes assigned manually. The clock bounding-box mismatch was corrected so Unity's shape check accepts the mesh.
- **Orbit ownership:** orbit replay uses its own registered reconstruction rather than another session's mesh.
- **Task shortcut:** every task object is currently served the latest mesh-bearing orbit registration, without checking object identity. Source orbit dimensions accompany the mesh; task tracks remain unchanged.
- **Matching disabled:** SigLIP retrieval achieved 4/11 correct historical decisions; correct and incorrect scores overlap. Automatic matching remains switched off.
- **Still needed:** validated identity and mesh-quality selection, mesh-backed pose estimation where appropriate, and headset checks of the latest API policy.

This patch joins display and storage; it does not make the borrowed mesh the right object or improve the recorded poses.

PointCloudPCA: pose fallback without a mesh

A partial view or noisy depth can make the estimated object position jump.

Module 5a

Partially fixed

This route estimates the centre and orientation of the visible depth points. If the visible patch changes, the estimated object can suddenly move even when the real object does not. The full object shape is unknown.

- | Slow candle: maximum frame-to-frame jump 0.153 m.
- | Fast clock: maximum frame-to-frame jump 1.053 m.
- | Table-contact Stanley cup: maximum jump 0.738 m.

These are recorded pre-guard measurements, not new results after the fix. During a confirmed grip, the deployed guard drops estimates more than 0.5 m from every available wrist/palm. Missing hand landmarks cannot be checked, and unheld objects are not covered by this rule.

Next: measure post-guard natural-motion performance and add justified motion, table-contact or temporal constraints. Robust fallback replay remains unconfirmed.

Automatic mesh matching

The identity matcher stays off while false assignments remain unresolved.

Module 7 **Disabled**

The full mesh-selection situation is collected in [Mesh selection and headset replay](#). This disabled branch is why normal live tasks do not automatically reach FoundationPose.

Replay orientation and interaction

Unity-side facing, placement, size and grabbing still need refinement.

Module 9 **Not confirmed**

- Align ghost replay with the viewer's current facing direction at spawn.
- Correct clock orbit mesh orientation.
- Define placement when new recording stamps are added, and check their sizing.
- Support both direct grabbing and Quest-style distance pointers.

These are reported follow-ups, not verified fixes. The latest mesh policy does not resolve them.

Dedicated Maestro server endpoint

The IT Services request is submitted; there has been no follow-up response.

Module 2 **Not confirmed**

Maestro currently uses the shared university proxy. A dedicated endpoint is pending for EEAI 2026 and the presentation of the encapsulated edge-AI system. A dedicated endpoint is the intended infrastructure change, but successful operation after migration is not yet established.

After IT supplies it, validate headset URLs, HTTPS, uploads, metadata and mesh downloads together; changing the endpoint may expose integration problems.

DINOv3 encoder consolidation

A single encoder could replace DINOv2 and SigLIP, but the corrected results do not justify adoption.

Module 7 **Not confirmed**

The aim is to simplify visual comparison and reduce model weight. DINOv3-S has 21.60M parameters and B 85.66M, versus 225.21M for the existing pair. The proposed replacement is not deployed.

Method	Same object across hands, at 0.82	Correct mesh decisions: historical 11 / all 15
Current DINOv2 + SigLIP	DINOv2: 3/4	SigLIP: 4/11 · 7/15
DINOv3-S	2/4	5/11 · 8/15
DINOv3-B	2/4	5/11 · 7/15
DINOv3-B foreground comparison	Not measured as a separate re-ID method	10/11 · 13/15; still two absent objects assigned incorrectly

The earlier test compared reused/self crops rather than independent hand-to-hand views, and some source sessions were empty or had wrong masks. That proxy could not establish real re-identification. The corrected evaluation restored the data and used independent crops; it is complete and supersedes that first test.

Foreground matching ranks known objects better, but still wrongly assigns an absent pen and remote. Next research needs independent held-out recordings, absent-object checks and reliable rejection of unknown objects; any adopted replacement must also preserve the identity pipeline's spatial/time checks. Keep DINOv2 and SigLIP, and leave mesh matching disabled meanwhile.

[Corrected evaluation and methods](#)

[DINOv3 overview and code](#)

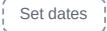
Possible feedback and design decisions

- **Naming convention:** should users keep the compact machine name, see a friendlier title, or add their own task title? Current names describe object records, not the action performed.
- **Offline Wi-Fi:** distinguish connected, unstable and fully offline states with an explicit icon/message. Agree how queued uploads and unavailable metadata should appear; client behaviour still needs validation.

Estimated Timeline

Envisioned · five full-time days per week across all routes. Click a coloured bar to expand its details. Drag it to move dates; drag its edges to change duration. Undated tasks can be scheduled from their boxes. Overlapping bars show planning windows, not separate full-time allocations. Viva scheduling may shift these estimates.

Solid: estimate Dashed: research / conditional Dark: fixed event / target Unscheduled: date to agree

Work route / task	Oct 2026	Nov	Dec	Jan 2027
Maestro VR				
Controlled task and capture protocol 5 Oct 2026 – 15 Oct 2026				
Orientation and replay refinement 22 Oct 2026 – 6 Nov 2026				
First controlled collection 9 Nov 2026 – 20 Nov 2026				
Collect in abundance and vet 23 Nov 2026 – 31 Dec 2026				
Cluster work				
Mesh identity and quality decisions 5 Oct 2026 – 15 Oct 2026				
Pose fallback and mesh selection 22 Oct 2026 – 6 Nov 2026				
OAK Lite + Vive enrichment research 26 Oct 2026 – 13 Nov 2026				
DINOV3 decision / bounded research 26 Oct 2026 – 6 Nov 2026				
Dataset export and quality review 9 Nov 2026 – 31 Dec 2026				
Kingswell case study 1 Dec 2026 – 31 Jan 2027				
Policy + world-model route Date to agree				
Dedicated endpoint / IT dependency Date to agree				
EEAI				
Presentation preparation 5 Oct 2026 – 15 Oct 2026				
Preparation / upload target 15 Oct 2026				
EEAI conference 19 Oct 2026 – 21 Oct 2026				
Master's viva				
Slides and question-bank preparation Date to agree				
Mock viva and revision Date to agree				
Viva Date to agree				

▼ Why the plan moved, and what remains to agree

The original technical plan expected regular expert recording from June, labelled data and training preparation in July–August, and initial robot simulation results by October 2026. Dataset construction depended on a reliable VR capture/processing tool. Mesh work took approximately one month according

to the project lead, followed by integration and tracking setbacks; this is not a measured breakdown of the whole delay.

The envisioned route is refinement → one controlled task collected in sufficient quantity and vetted → Kingswell case study → further collection and robotics inference. Sample count, coverage, quality thresholds, hardware integration and viva scheduling remain to agree. Hardware research is optional unless the agreed collection criteria require it.

Sources: technical delivery plan; EEAI acceptance letter; project audit evidence; Master's viva preparation notes. No live presentation or demo requirement is assumed.

Timeline entry notes

Maestro VR

Controlled task and capture protocol

5 Oct 2026 – 15 Oct 2026

Agree one small task, required streams, annotations, sample-count and coverage targets.

Ready when: The collection and acceptance rules are written down.

Orientation and replay refinement

22 Oct 2026 – 6 Nov 2026

Check mesh orientation, ghost-hand facing, recording placement/size and direct or distance grabbing.

Ready when: Fresh recordings pass headset checks; remaining limitations are visible.

Depends on: Unity availability and headset validation.

First controlled collection

9 Nov 2026 – 20 Nov 2026

Repeat the agreed task and review the initial batch before increasing volume.

Ready when: A usable initial batch with synchronised data and complete metadata.

Depends on: Capture and cluster quality checks.

Collect in abundance and vet

23 Nov 2026 – 31 Dec 2026

Continue the same task with agreed variations; review hand confidence, masks, object poses, labels and data splits.

Ready when: The agreed quantity, coverage and quality criteria are met. This window extends if they are not.

Depends on: Accepted initial batch; thresholds and sample count still to agree.

Cluster work

Mesh identity and quality decisions

5 Oct 2026 – 15 Oct 2026

Establish identity before choosing among that object's mesh candidates. Define scale, completeness, alignment and tracking checks.

Ready when: A justified selection and quality policy replaces the latest-orbit shortcut for collection.

Pose fallback and mesh selection

22 Oct 2026 – 6 Nov 2026

Implement the agreed mesh policy and validate the existing PCA guard. Separate usable estimates, manual-review cases and rejected frames.

Ready when: The live upload-to-processing path passes fresh controlled trials.

Depends on: Agreed mesh policy; enough recordings to measure failure cases.

OAK Lite + Vive enrichment research

26 Oct 2026 – 13 Nov 2026

Investigate an external view and a tracked reference for the controlled task. If the camera is OAK-D Lite, RGB and stereo depth could add views of occluded actions; a rigidly mounted Vive tracker could provide a separate pose reference. Confirm the exact devices and tracking setup first.

Ready when: A short feasibility study compares Quest-only recordings with added sensors: missing frames, time alignment, pose discrepancies and setup effort. Integration is justified by measured benefit.

Depends on: Device model/setup confirmation, timestamp synchronisation, camera calibration, Quest/tracker coordinate alignment and tracker-to-object offset. Extra hardware is not automatically ground truth. [Luxonis depth overview](#) · [Vive tracking-system comparison](#)

DINOv3 decision / bounded research

26 Oct 2026 – 6 Nov 2026

The corrected evaluation is already complete and does not justify adoption. Keep DINOv2 and SigLIP; only run additional research against a defined new question.

Ready when: Retain the current pair or justify replacement with independent positive and absent-object checks.

Depends on: Optional research; collection need not wait if its identity workflow is validated.

Dataset export and quality review

9 Nov 2026 – 31 Dec 2026

Vet synchronisation, fields, confidence, identity, masks, tracks, annotations, coverage and train/validation/test separation. Add sensor streams only if the enrichment trial warrants them.

Ready when: A sufficiently large vetted dataset is accepted for the controlled task.

Depends on: Agreed criteria and collected recordings.

Kingswell case study

1 Dec 2026 – 31 Jan 2027

Use the accepted dataset to agree what to collect next and what the chosen robot can infer or execute. December–January is an indicative discussion window, not a promised start.

Ready when: Robot/task scope, observations/actions, coordinate mapping, success measures and execution environment are agreed.

Depends on: Dataset acceptance FIRST and Kingswell availability.

Policy + world-model route

Date to agree

After the case-study review, scope a baseline robot policy and a world-model route informed by the Master's research. DiWA is the likely research starting point; human wrist recordings do not directly supply robot controls. [DiWA overview and code](#).

Ready when: A feasible data/interface plan and effort estimate, then task-success evaluation in simulation.

Depends on: Kingswell review, robot interaction data, success signal and execution checks.

Dedicated endpoint / IT dependency

Date to agree

The IT request is submitted. Schedule migration only once an endpoint is supplied, and check uploads, metadata and mesh downloads together.

Ready when: End-to-end validation on the supplied endpoint.

Depends on: IT response; timing outside project control.

EEAI

Presentation preparation

5 Oct 2026 – 15 Oct 2026

Explain headset sensing/local tracking/rendering versus heavy AI processing on the cluster. Presentation format and any demonstration remain open for discussion.

Ready when: Material ready for upload before 16 October, as requested in the acceptance letter.

Depends on: Organiser format confirmation; no live headset demo assumed.

Preparation / upload target

15 Oct 2026

15 October is the internal target for the stated before-16-October deadline.

Ready when: Presentation material uploaded using the organiser's instructions.

EEAI conference

19 Oct 2026 – 21 Oct 2026

Conference participation reduces available development time. [Conference programme](#).

Ready when: Participation and presentation arrangements confirmed.

Master's viva

Slides and question-bank preparation

Date to agree

Prepare a concise account of the submitted research, results, limitations and likely examiner questions. The recorded working aim is about 15 slides / 20 minutes; the chair's confirmed format takes precedence.

Ready when: Slides and a reviewed answer bank match the submitted thesis.

Depends on: Confirm viva date and official format; reserve time within the same five-day working week.

Mock viva and revision

Date to agree

Rehearse with colleagues and the supervisor, then refine explanations and timing.

Ready when: A timed rehearsal and reviewed responses to difficult questions.

Depends on: Preparation material and participants' availability.

Viva

Date to agree

No confirmed viva date is recorded. Schedule preparation backwards from the official date once it is known.

Ready when: Official date and format confirmed; preparation time accounted for.

Depends on: Board/chair notice.

Technical Detail

[Full technical audit and measurements](#)

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